LDL × CAD across three levels of QC

One trait pair, three cleaning stages, two platforms.
bipred 0.3.5 (macOS/arm64) and 0.3.6 (Windows/x86-64), ldpred3 0.4.5.
Estimates in `results/estimates.csv`.

The question

The simulation benchmarks draw $\hat{\beta} \sim N(R\beta, R/N)$ from the model the sampler assumes. That is the right way to measure an estimator against a known truth, and it is structurally incapable of catching a failure that appears only when the summary statistics disagree with the LD reference — which is exactly what shipped in 0.3.0, where the first real GWAS bipred was pointed at produced a silently diverged fit that thirty architecture cells had no way to detect.

This study asks what three levels of cleaning do to one real bivariate fit: harmonisation alone, per-variant filters, and those filters followed by the LD-consistency screen.

Traits

LDL — GLGC 2013 (Willer et al.), continuous, per-variant N.

CAD — CARDIoGRAMplusC4D 2015 (Nikpay et al.), GWAS Catalog GCST003116. Case/control, so fitted at an effective N. Its standard-error column is genomic-control corrected, which deflates z-scores, so the CAD heritability reported here is observed-scale at the study's case fraction *and* conservative. It is not a liability-scale heritability.

The two consortia are believed close to disjoint. Their cross-trait LDSC intercept ($\sim +0.02$) is consistent with small correlated sampling error, but an intercept cannot identify cohort overlap by itself.

The screen is the load-bearing step

Of the three cleaning levels, only the last produces a fit that is admissible at all.

Cancellation — $\sum \beta^2/h^2$, the ratio that detects effects adding up through LD rather than to the heritability they are supposed to explain — starts at 255 for LDL and stays there through per-variant filtering, reaching 271. Minor allele frequency, imputation quality and a χ^2 cap remove 37,000 variants and leave the pathology untouched, because none of them can see it: each judges a variant on its own. The screen drops a further 42,000 and cancellation falls to 0.65.

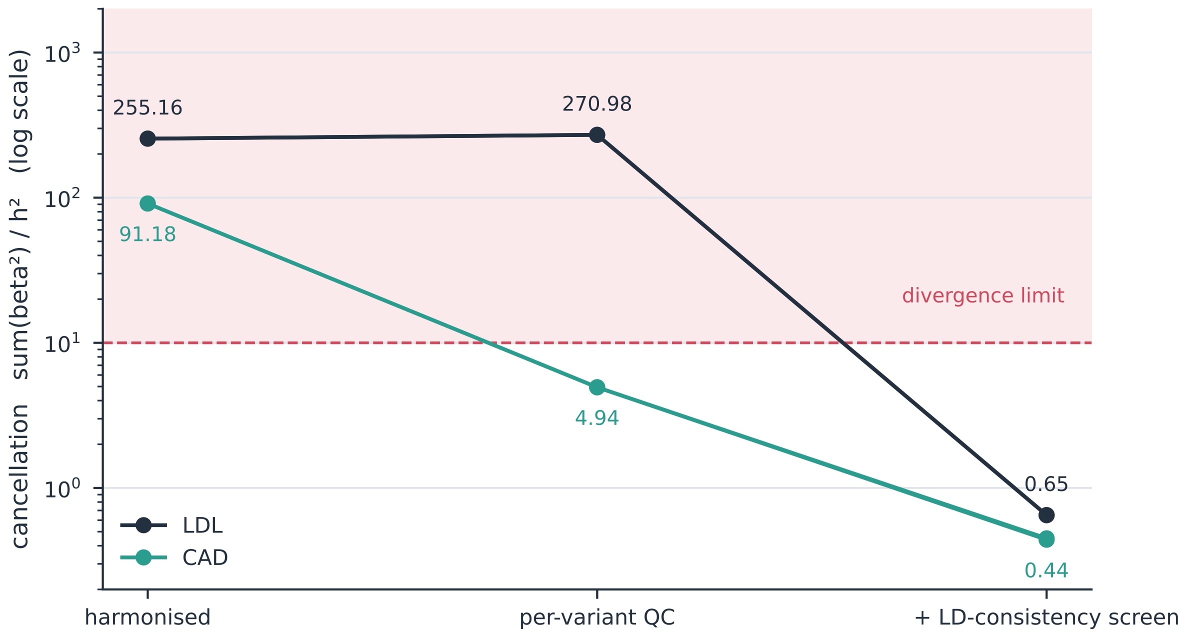

Figure 1: Cancellation by cleaning stage, log scale. Above the limit of 10 the sampler is placing large opposing effects on variants in near-perfect LD, and the heritability it reports is those effects cancelling rather than explaining. Per-variant filters move LDL the wrong way. CAD crosses the limit at the per-variant stage but only clears it after the screen. Both platform runs are drawn; they coincide closely enough to be one line each.

Only after the screen do all three divergence diagnostics clear. That is what makes the third row an estimate and the first two not.

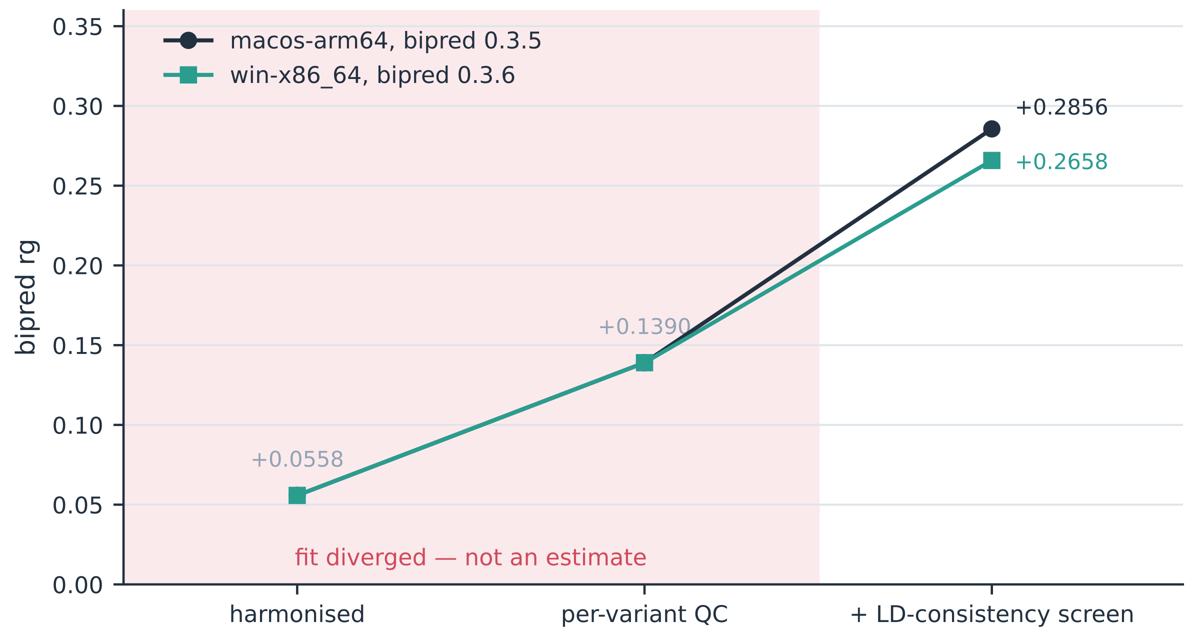

Figure 2: Genetic correlation by cleaning stage. The shaded region marks the stages where the fit carried a divergence warning: the +0.0558 and +0.139 plotted there are what a diverged sampler emitted, not estimates of anything, and the apparent trend across the three points is not a convergence toward a right answer. The $\sim +0.27$ that emerges is a consequence of the fit becoming valid.

Heritability moves with it. LDL's h^2 falls from 0.67 to 0.09 across the three stages, which is the same phenomenon read on a different scale: the 0.67 was cancellation, not signal.

This is one trait pair. It is why the screen exists and why arm 001 fixes it on; it does not establish a universal QC rule, and the $2 \times 2 \times 2$ factorial that would support one is in `benchmarks/qc_factorial.py`.

A caution this study exists to make

`results/estimates.csv` carries two runs of the same three stages: macOS/arm64 under bipred 0.3.5, and Windows/x86-64 under 0.3.6, against an independently rebuilt LD reference.

Stages 1 and 2 agree to every recorded digit — across a different operating system, architecture and BLAS. Stage 3 does not: `rg` moved from +0.2856 to +0.2658, and m from 845,623 to 845,423.

The cause is benign. Version 0.3.6 deliberately gave up reproducing 0.3.5's screening masks for a given seed, so about 200 of 845,000 variants differ. But the shape of it is the reason `studies/` is a separate directory from `benchmarks/`: **a biological estimate moved because of an implementation refactor**, and under a regenerate-every-release contract nothing would have flagged it.

That is also the honest scale for the third-stage number. A 0.02 shift from reseeded masks is the resolution this pair supports; +0.27 is the result, not +0.2658.

Methods

The model

bipred is bivariate LDpred: one joint fit of two traits that share a single LD reference. Each variant is assigned to one of four latent states — causal for neither trait, for trait 1 only, for trait 2 only, or for *both* — with probabilities $(\pi_{00}, \pi_{10}, \pi_{01}, \pi_{11})$. A trait-1-only effect is drawn from $N(0, s_1)$, a trait-2-only effect from $N(0, s_2)$, and a both-causal pair from $N(0, \Sigma)$ with $\Sigma = \left[\begin{smallmatrix} s_1 & s_{12} \\ s_{12} & s_2 \end{smallmatrix}\right]$. The off-diagonal s_{12} is the effect covariance inside the shared component, and is the only place the two traits couple.

The indicator is *per trait*, not a single shared one. Whether the two traits' causal variants co-occur is therefore learned as π_{11} rather than assumed: correlated traits borrow strength through the both-causal component, and disjoint traits can drive π_{11} toward zero so the two fits decouple. That is what makes the polygenic-overlap summary an estimate rather than a restatement of the prior.

Fitting is Gibbs sampling over the LD blocks. Each sweep evaluates the four bivariate-Gaussian likelihoods of the residual marginal estimate at a variant, samples that variant's state, then draws its effects; the mixture weights π and the variance components (s_1, s_2, s_{12}) are re-estimated every sweep under a damping factor of 0.2. Reported quantities are posterior means over the retained iterations: per-trait SNP heritability `h2`, the genetic correlation `rg`, posterior-mean effects, and a MiXeR-style overlap summary — `n_causal` per trait, `n_shared`, `frac_shared`, and the correlation `rho_beta` of effects within the shared component.

What rho_beta is, and is not

`rg` is a genome-wide average: genetic covariance divided by the geometric mean of the two heritabilities, over every variant in the fit. `rho_beta` is the effect correlation *conditional on a variant being causal for both traits.* The two answer different questions, and they come apart exactly when one trait's heritability sits in very few loci. Neither is a causal estimate.

`frac_shared` is π_{11} divided by the smaller of the two marginal causal fractions, so it asks what proportion of the less polygenic trait's causal set is also causal for the other.

Sample overlap

Overlapping samples correlate the two traits' sampling errors. The fit accepts that correlation as `cross_corr`, set per pair from the cross-trait LD Score regression intercept, used as-is and never clipped into range. An intercept can also reflect correlated confounding rather than shared samples, so this absorbs overlap without identifying it.

The LD-consistency screen

Per-variant filters — minor allele frequency, imputation quality, a chi-square cap, sample size — judge each variant in isolation, so none of them can see a variant disagreeing with the variants it is correlated with. The screen tests exactly that. Within a window the variants are split at random in half and each z-score in one half is predicted from the other through the LD,

$$\hat{z}_a = R_{aB}\, R_{BB}^{+}\, z_B, \qquad T_a = \frac{(z_a - \hat{z}_a)^2}{1 - R_{aB}\, R_{BB}^{+}\, R_{Ba}} \;\sim\; \chi^2_1$$

under consistency. Variants far from what their neighbours predict are dropped, and the split is repeated so each variant is tested from several directions.

It runs against the same numeric LD representation the fit will use, so the screen and the sampler see the same matrix. It is DENTIST-inspired, not DENTIST: window construction, partition schedule, eigenvalue regularisation and removal policy all differ from the published pipeline (Chen et al. 2021, *Nature Communications* 12:7117), so that pipeline's calibration does not transfer.

Divergence diagnostics

A bivariate fit can diverge silently. Given summary statistics that disagree with the LD reference, the sampler places large opposing effects on variants in near-perfect LD, and these cancel to a plausible-looking `h2`. Three diagnostics are recorded, and any one of them raises the warning: $\sum \beta^2/h^2$ above 10 (the cancellation ratio, reported in both studies), $\max|\beta|$ above 25 slab standard deviations, and retained-trace drift above 1.25. All three were calibrated on one real LDL $\times$ CAD fit that diverged and the same fit after screening repaired it, so each separates the two regimes by more than an order of magnitude rather than by a hair.

An estimate carrying the divergence warning is not an estimate of anything, and is reported only to show the contrast.

LD reference and harmonisation

All fits use the European UK Biobank HapMap3 LD reference, built with ldpred3's bigsnpr converter at the pinned revision `5d86ac9d`. Summary statistics are harmonised to it by rsID with allele matching, flipping the effect sign where the effect and reference alleles are swapped; strand-ambiguous and unmatched variants are dropped. Odds-ratio releases are taken to logs before harmonisation, which preserves the sign across a flip. Case/control traits are fitted at an effective N of $4/(1/n_{\text{case}} + 1/n_{\text{ctrl}})$. BLAS is pinned to one thread throughout.

This analysis

Three cleaning stages are fitted, and their contrast is the result:

1. **harmonised** — rsID match to the LD reference with allele matching and sign flipping only. 924,254 variants.
2. **per-variant QC** — adds MAF ≥ 0.01, imputation INFO ≥ 0.9, $\chi^2 \leq 80$, and allele-frequency discordance against the reference ≤ 0.20. 887,361 variants.
3. **+ LD-consistency screen** — adds the screen of the previous section at four rounds. 845,623 variants (845,423 on the second run).

Each stage is a full joint fit at 200 burn-in and 300 retained sweeps, with `cross_corr` taken from that stage's own cross-trait LDSC intercept. Long-range LD regions are retained; the MHC is removed with the stage-2 filters.

The chi-square cap applies here too

The $\chi^2 \leq 80$ term enters at stage 2 and carries into stage 3, so two of the three rows are fitted on a capped variant set. The cap is inherited from LD Score regression, whose linear model gives an uncapped large-effect variant near-full leverage on the slope; the bivariate fit does not need it and gets it anyway, because both estimators are handed one variant set.

For LDL and CAD the cost should be small — both are polygenic, and no single locus carries a large share of either. It is not zero: the cap is what removes the *LPA* region from any lipid trait, and for CAD it removes the strongest signals at 9p21. The companion `oligogenic-overlap` study documents the filter in detail, where for lipoprotein(a) it removes 73% of the causal locus and 42.5% of the trait's summed chi-square.

What can be said from the numbers here is narrower: the cap did not repair the divergence. Cancellation for LDL is 255 before it and 271 after. Whatever the cap is doing at stage 2, it is not what makes the fit admissible — that is the screen.

The generating script is `benchmarks/real_ldl_cad.py`, which stays in `benchmarks/` because its other role — does a real-shaped GWAS still produce an admissible fit? — is a regression canary that should be regenerated freely. What belongs here is the estimate and its reading. The two roles want opposite handling, which is why they live in different places.

Inputs are not committed; `benchmarks/README.md` Table 4 records how to acquire them, and `../_lib/traits.toml` registers their column layouts.